

Agronomy for Sustainable Development

Soil-health frameworks in agri-food systems. A review

--Manuscript Draft--

Manuscript Number:	ASDE-D-25-01749R4	
Full Title:	Soil-health frameworks in agri-food systems. A review	
Article Type:	Review Article	
Keywords:	Soil Health; Agroecology; Agri-Food Systems; framework; program design; implementation; Evaluation; sustainable development.	
Corresponding Author:	Powell Mponela NEPAL	
Corresponding Author Secondary Information:		
Corresponding Author's Institution:		
Corresponding Author's Secondary Institution:		
First Author:	Powell Mponela	
First Author Secondary Information:		
Order of Authors:	Powell Mponela	
	Vimbayi Grace Petrova Chimonyo	
	Mazvita Chiduwa	
	Righteous Kachali	
	Joyce Grevulo-Minofu	
	Cleopatra Kawanga	
	Santiago Lopez-Ridaura	
	Sieglinde Snapp	
Order of Authors Secondary Information:		
Funding Information:	Consortium of International Agricultural Research Centers	Prof Sieglinde Snapp
	MwAPATA Institute	Ms Joyce Grevulo-Minofu
	McKnight Foundation	Prof Sieglinde Snapp
Abstract:	Soil health is central to agroecological transitions, yet guidance for integrating it into agri-food system design and monitoring remains fragmented. Institutions increasingly use frameworks to define indicators, guide interventions, and report progress against climate, biodiversity, and food-security agendas. However, to our knowledge, there is no integrative soil health framework which coherently links biophysical diagnostics, socio-institutional enablers, and multiscale accountability. This leaves critical gaps in design, sequencing, and measurement of agroecological transitions. Here we review how soil health is operationalized within agroecology and agri-food systems and translate these patterns into an actionable programming guide. We reviewed 64 frameworks and extracted 652 indicators across 12 agroecological principles to build a framework-by-principle evidence matrix. Frameworks were classified by use-orientation (theory, practice, analysis), and indicator thematic profiles were analyzed using hierarchical clustering with adaptive branch detection. The major findings are as follows: 1) frameworks evolution exhibits four chronological waves with shifts from conceptual foundations to operational measurement and outcome reporting, alongside changes in global and regional agenda setting and a rising demand for comparable indicators; 2) clustering identified five soil health design	

	domains separating biophysical and socio-economic principles and revealing stable micro-constellations beyond earlier pathway framing. These include soil management and input stewardship, soil-health assessment, agroecological and ecosystem-based, integrated landscape and livelihood, and policy- and outcome-based. These findings were translated into a sequenced, multi-domain programming architecture that operationalizes complementarity across diagnostics, stewardship implementation, ecosystem safeguards, landscape–livelihood embedding, and iterative learning, thereby closing the gaps between farm practices, governance mechanisms, and outcome monitoring for soil health.
Author Comments:	

Managing Editor,
Agronomy for Sustainable Development

Submission of a revised manuscript (ASDE-D-25-01749R4)

Dear Managing Editor Béatrice Trotier,

Thank you for your constructive feedback, with refinements.

I have abbreviated the remaining journal names following the nomenclature available in Mendeley – aligned with ASD and also adjusted the labels for Fig 2.

Kind regards,

Powell Mponela

On behalf of co-authors.

Soil-health frameworks in agri-food systems. A review

Powell Mponela^{1@}, Vimbayi Grace Petrova Chimonyo^{2,3}, Mazvita Chiduwa^{2,4}, Righteous Kachali^{4,5}, Joyce Grevulo-Minofu⁶, Cleopatra Kawanga⁷, Santiago Lopez-Ridaura⁸, Sieglinde Snapp⁸

¹CIMMYT, NARC Khumaltar, Lalitpur, P.O. Box: 5186. Kathmandu, Nepal

²CIMMYT, 12.5km Peg, Mazowe Road, Mount Pleasant Harare, Zimbabwe

³Centre for Transformative Agriculture and Food Systems, University of Kwazul-Natal, Scottsville, Pietermaritzburg, South Africa

⁴CIMMYT, P.O. Box 1096, Chitedze Research Station, Mchinji Road, Lilongwe, Malawi

⁵Lilongwe University of Agriculture and Natural Resources (LUANAR), Bunda Campus, P.O. Box 219, Lilongwe, Malawi.

⁶Mwapata Institute, Lundu Street, Area 10/386, P.O. Box 30883, Capital City, Lilongwe, Malawi

⁷CIMMYT, Plot # 4108, Mwinilunga Road, Sunningdale, Lusaka - Zambia

⁸CIMMYT, Carretera Mexico-Veracruz, Km. 45, El Batan, 56237 Texcoco, Mexico

Corresponding author: Powell Mponela, p.mponela@cgiar.org

Abstract

Soil health is central to agroecological transitions, yet guidance for integrating it into agri-food system design and monitoring remains fragmented. Institutions increasingly use frameworks to define indicators, guide interventions, and report progress against climate, biodiversity, and food-security agendas. However, to our knowledge, there is no integrative soil health framework which coherently links biophysical diagnostics, socio-institutional enablers, and multiscale accountability. This leaves critical gaps in design, sequencing, and measurement of agroecological transitions.

Here we review how soil health is operationalized within agroecology and agri-food systems and translate these patterns into an actionable programming guide. We reviewed 64 frameworks and extracted 652 indicators across 12 agroecological principles to build a framework-by-principle evidence matrix. Frameworks were classified by use-orientation (theory, practice, analysis), and indicator thematic profiles were analyzed using hierarchical clustering with adaptive branch detection.

The major findings are as follows: 1) framework evolution exhibits four chronological waves with shifts from conceptual foundations to operational measurement and outcome reporting, alongside changes in global and regional agenda setting and a rising demand for comparable indicators; 2) clustering identified five soil health design domains separating biophysical and socio-economic principles and revealing stable micro-constellations beyond earlier pathway framing. These include soil management and input stewardship, soil-health assessment, agroecological and ecosystem-based, integrated landscape and livelihood, and policy-and outcome-based. These findings were translated into a sequenced, multi-domain programming architecture that operationalizes complementarity across diagnostics, stewardship implementation, ecosystem safeguards, landscape–livelihood embedding, and iterative learning, thereby closing the gaps between farm practices, governance mechanisms, and outcome monitoring for soil health.

Keywords: soil health; agroecology; agri-food systems; framework; program design; implementation; evaluation; sustainable development.

Contents

40			
41	1	Introduction	2
42	2	Framing soil health within Agroecological and agri-food systems	4
43	2.1	Framework definition and institutional strategies	4
44	2.2	Principles and indicator families used to represent soil health in agroecology.....	5
45	2.3	Review scope and synthesis approach.....	5
46	3	Framework evolution	6
47	3.1	Use-orientation	7
48	3.2	Agroecological principle-orientation	9
49	4	Design domains.....	11
50	4.1	Soil management and input stewardship	12
51	4.2	Soil-Health Assessment	12
52	4.3	Agroecological and Ecosystem-Based	13
53	4.4	Integrated Landscape and Livelihood	14
54	4.5	Policy and outcome-orientation.....	15
55	5	Integration and programming	15
56	6	Conclusion	17
57	7	Acknowledgements	18
58	8	Declarations.....	18
59	8.1	Funding	18
60	8.2	Conflict of interest	18
61	8.3	Ethics and consent.....	18
62	8.4	Availability of data and material	19
63	8.5	Code availability	19
64	8.6	Authors contributions	19
65	9	References	19
66			

1 Introduction

The main challenge in agroecological food systems is to balance feeding a growing population while safeguarding environmental integrity (Pinstrup-Andersen and Hazell 1985; Rockström et al. 2009; Godfray et al. 2010; Foley et al. 2011). Soil health is central to this challenge because it underpins agroecosystem productivity and resilience through water and nutrient regulation and carbon storage, with direct implications for climate mitigation, food security, biodiversity conservation, and environmental sustainability (Lal 2004; Kibblewhite et al. 2008; Amundson et al. 2015; IPCC 2019; Madsen et al. 2025). Yet soil degradation now affects over a third of the global agricultural land, driven largely by unsustainable practices, deforestation, and urbanization (FAO-ITPS 2015; IPCC 2019). Beyond biophysical functions, soil health is also foundational to One Health and Planetary Health, including issues such as antimicrobial resistance, zoonotic disease emergence, and terrestrial pollution (Montgomery et al. 2024). These pressures are especially consequential in regions where agriculture underpins livelihoods for most of the population; declining soil health reduces productivity, deepens food insecurity, and increases vulnerability to climate shocks (Sanchez 2002; Amundson et al. 2015; IPCC 2019).

Although the science of soil processes is well established (Figure 1), evidence and guidance are fragmented and insufficiently contextualized to local socio-ecological realities, limiting their translation into actionable interventions and policy design. Science-based frameworks in agroecological and agri-food system transitions can guide planning, implementation, and evaluation of soil health, but persistent gaps remain in comprehensiveness, inclusivity, and applicability across diverse farming systems (Cowie et al. 2018; Wezel et al. 2020; Utter et al. 2021). While frameworks provide a shared language for stakeholders, their proliferation has generated inconsistencies in scope and design, hindering their collective utility (Hazard et al. 2026). Without coherent integration across biophysical, socio-economic, and governance dimensions, soil health is often monitored as a technical variable rather than being embedded as a strategic design objective within transition processes. As a result, interventions risk remaining project-bound and episodic, limiting cumulative learning, cross-scale coordination, and accountability for long-term soil stewardship outcomes.



Figure 1 Measuring soil infiltration for land degradation surveillance at Linthipe Dedza District in Malawi (Photocredit: Powell Mponela, International Center for Tropical Agriculture (CIAT)).

Soil health assessment tools offer operational benchmarks but are frequently applied in isolation from broader sustainability planning (Moebius-Clune et al. 2017; USDA-NRCS 2021). Conversely, systems-oriented frameworks increasingly integrate biodiversity, water, livelihoods, climate, and governance, yet often treat soil health indirectly - as a secondary outcome of land-use planning, ecosystem service delivery, or sustainability transitions rather than as a primary design objective (Sanchez 2002; Sayer et al. 2013). For example, FAO's voluntary guidelines provide global practice standards but limited granularity for heterogeneous landscapes (FAO-ITPS 2015, 2017); surveillance approaches can prioritize biophysical metrics while under-addressing socio-economic barriers such as gendered land tenure (Amundson et al. 2015; Utter et al. 2021); and carbon-oriented initiatives can prioritize climate and planetary health benefits while missing near-term agronomic decision needs unless coupled with decision support (Lal 2004; Minasny et al. 2017; Montgomery et al. 2024). These limitations point to a persistent gap between conceptual integration and practical transformation, particularly in dynamic smallholder contexts where scaling from fields to landscapes depends on governance, knowledge systems, and learning under climate and market volatility (Foley et al. 2011; Sayer et al. 2013; Cowie et al. 2018; IPCC 2019).

Agroecology offers a systems lens for repositioning soil health within wider agri-food systems by emphasizing interdependencies among ecological processes, livelihoods, and institutions, and highlighting diversity, resilience, resource efficiency, and the co-creation of knowledge (HLPE 2019; Nicholls et al. 2020; Wezel et al. 2020; Dumont et al. 2021). In this framing, soil health is not only a determinant of production but also a provider of ecosystem services and a foundation for justice, governance, and community connectivity (Sayer et al. 2013; Utter et al. 2021). For instance, for smallholders in fragile environments with limited external inputs, agroecological practices — including nutrient recycling, diversified rotations, and organic amendments — offer practical pathways to regenerate soils and strengthen climate resilience (Sanchez 2002; Tittonell 2014; Altieri et al. 2015; Madsen et al. 2025). The growing literature also presents methodological challenges, as stakeholders struggle to align the perspectives requiring knowledge co-creation and collective learning (Andrieu et al. 2025; Mockshell et al. 2026).

Against this backdrop, the main objective of this review is to examine how soil health is embedded within agroecological and agri-food system framework designs. Specifically, we (i) map how soil health indicators align with the ecological, social, cultural, and economic dimensions articulated by agroecological principles; (ii) consolidate existing frameworks into a comparable evidence matrix to identify design domains; and (iii) translate these domains into a sequenced programming guide that operationalizes their complementarity.

2 Framing soil health within Agroecological and agri-food systems

Frameworks that address soil health in agroecology and agri-food systems differ widely in purpose, scope, and evidence they provide. This section attempts to establish a common understanding - definitions, principle structure, and synthesis - so that heterogeneous framework families can be compared on a consistent basis and interpreted through the framework-by-principle evidence matrix and the design domains developed in sections 3 and 4.

2.1 Framework definition and institutional strategies

The study adopts the dictionary definition of a framework as “a supporting structure around which something can be built: a system of rules, ideas, or beliefs that is used to plan or decide on something” (Cambridge University Press 2025). The diversity of frameworks observed in agri-food, agroecology, and soil health literature reflects more than disciplinary or methodological differences; it arises from distinct institutional logics and normative orientations that guide how problems are framed, and how solutions are designed. Institutions aim to maximize collective benefit while minimizing costs through frameworks that guide the design of context-relevant interventions (Ostrom 2011). From this perspective, different frameworks emerge and represent competing bundles of theories, practices, and analysis indicators. Institutions are assumed to behave rationally by formulating or selecting the framework that yields the highest utility in terms of policy coherence, operational feasibility, and alignment with its technical, societal, and environmental goals (Aleskerov et al. 2007; Moullin et al. 2020). However, where indifference exists, multiple frameworks may be treated as functionally equivalent or complementary.

Notwithstanding, the utility-maximization process is not value-neutral. Institutional decisions are also shaped by embedded belief systems and normative commitments, as described by the Value-Belief-Norm (VBN) theory of environmentalism (Oreg and Katz-Gerro 2006). The VBN posits that values influence environmental beliefs, which in turn shape behavioral norms and decisions - such as selecting frameworks that prioritize equity, ecosystem integrity, or inclusive governance. The convergence of utilitarian logic and normative values forms the basis for institution's framework selection, shaping how agroecological transitions are designed, implemented and analyzed. This theoretical framing provides the basis for interpreting patterns observed in the evidence matrix. It reveals that the diversity of frameworks and the uneven representation of agroecological principles is not merely methodological but reflects distinct institutional strategies shaped by both utilitarian reasoning and normative commitments. Framework diversity is not only technical; it reflects institutional strategies shaped by utility-maximizing choices and normative commitments that determine which problems, indicators, and interventions are prioritized. Making these drivers explicit is essential for appraising frameworks in ways that balance operational feasibility with social legitimacy and contextual relevance.

2.2 Principles and indicator families used to represent soil health in agroecology

To compare frameworks consistently, the review is organized around agroecological principles and associated indicator families, treating soil health as a key, measurable component that interfaces with other principles (e.g., nutrient cycling, biodiversity, governance, and livelihoods). The principle-based structure primarily draws from agroecological principles and elements (HLPE 2019; Wezel et al. 2020), the tool for agroecology performance evaluation (TAPE) (FAO 2019; Mottet et al. 2020), and the characterization of agroecology transformation (CAET) procedure (Mottet et al. 2020; Lucantoni et al. 2023; Bicksler et al. 2023).

Indicator families were refined using supporting literature that elaborates specific principle areas – including co-creation of knowledge (Utter et al. 2021), human and social values (Bezner Kerr et al. 2022), diversity (Nicholls et al. 2020), animal health (Dumont et al. 2024; Ceppatelli et al. 2025) and soil health measurement guidance (Moebius-Clune et al. 2017; Apfelbaum et al. 2019; FAO-OEWG 2022; Montgomery et al. 2024). This approach enabled a consistent comparison of how frameworks translate principles into measurable priorities. Organizing the review around agroecological principles and indicator families provides a transparent, comparable basis for identifying where soil health is directly operationalized versus indirectly assumed within the broader ecological and socio-institutional priorities.

2.3 Review scope and synthesis approach

Literature and grey sources were compiled through structured searching and iterative refinement of keywords to capture diverse framework families spanning agriculture, ecology, and environmental governance. For each framework, indicators were extracted and mapped to the agroecological principle

structure (including soil health) to build an evidence matrix used to support both descriptive synthesis and the domain-oriented interpretation.

A Boolean document search was conducted between October 2024 and January 2025 to provide an initial mapping of relevant frameworks. The review was designed to synthesize framework designs (and their indicator emphases) using quantitative structuring (e.g., indicator extraction and clustering) while avoiding statistical inference. To capture frameworks across agriculture, ecology, and environmental sectors, the guiding query combined terms for the focal theme, framework type, and related paradigms: (soil health OR landscape) AND (framework OR analytical framework OR conceptual framework OR theoretical framework OR empirical framework) AND (agroecology OR sustainable intensification OR regenerative agriculture OR conservation agriculture). Searches were conducted in Google Scholar, Scopus, and institutional repositories; query syntax was adapted to platform-specific requirements.

As the compendium expanded, the search was iteratively extended using additional keywords to capture emerging terminology and related framework families. While the full sequence of evolving search strings was not recorded, the Boolean structure above reflects the guiding formulation used throughout the review and provided the basis for mapping framework diversity and situating soil health within wider systems-oriented approaches. Indicators were extracted from documents using the R package pdftools and grouped under the 13 agroecological principles (including soil health). Indicator context was captured by extracting proximal text (extended sentences) around each indicator to support consistent interpretation during mapping and to inform interpretation where principle boundaries are contested.

3 Framework evolution

The review uses two complementary lenses to interpret framework diversity. First, frameworks are classified by use-orientation based on established criteria (Rocco and Plakhotnik 2009; UNHCR 2017; Luft et al. 2022): (i) theory-oriented frameworks synthesize concepts to build foundations for theory development; (ii) practice-oriented frameworks conceptualize practices and their associated impacts; and (iii) analysis-oriented ones that provide structured approaches for gathering, analyzing, and interpreting data. This classification clarifies a framework's primary purpose and how it is intended to be used across program cycles.

Beyond use-orientation, the study examines how frameworks differ in their indicator emphasis across the agroecological principle structure (including soil health). To quantify and compare the representation of agroecological principles across diverse frameworks, composite indices were constructed (Nardo et al. 2005) by first applying z-score standardization to indicator counts, followed by unweighted aggregation (mean) at agroecology principle-level. The constructs were specified reflectively, if observed indicators manifest underlying soil health and agroecological principles (MacKenzie et al. 2011). The study used aggregation and averaging rather than dimensionality reduction methods (e.g. PCA or factor analysis) because these techniques can reduce interpretability and impose statistical assumptions (e.g.,

orthogonality, linearity) that may not align with multidimensional sustainability constructs (Greco et al. 2019). Aggregation preserves a transparent, additive structure aligned with the conceptual design, supporting clarity for stakeholders and decision-makers (Mazziotta and Pareto 2016).

To interpret framework diversity through pattern analysis, the study applied a row–column hierarchical clustering (simultaneous clustering of frameworks and principles). Clusters were identified using the Dynamic Tree Cut method in R - an adaptive branch-detection algorithm applied to the hierarchical clustering dendrogram that delineates clusters from dendrogram topology and branch stability, rather than imposing a single fixed cut height (Langfelder et al. 2008). The clustering was not used to rank or evaluate frameworks, it summarizes similarities and differences in indicator emphasis to help users identify comparable measurement priorities and assemble complementary framework portfolios that reduce thematic gaps across soil health, agroecology, and agri-food systems, while providing a shared reference for stakeholder communication.

3.1 Use-orientation

The review reveals distinctive use-orientations and evolutions of frameworks to support practical selection and adaptation, than prescribing a universal approach. From the compendium of 64 agro-food system, soil health and related frameworks, (Figure 2a), almost half focused explicitly on informing either the theory, practice or analysis, one-fourth focused on two dimensions and one-tenth, on all three. Higher coverage of theories at 68% and analytics at 60 %, while only 36 % focus on practices is observed. The explicit focus on theory and analytics signals increased intents to analyze linkages between organizational objectives, design concepts and outcomes, although mere indicator presence may not be related to measurement quality. There is a debate on whether the institutional and program design should focus on the outcomes, interventions, principles or all. Although the common approach has been to prioritize specific interventions (Derpsch 2003; Vanlauwe et al. 2014; Giller et al. 2015; Thierfelder et al. 2018) or adhering to core principles such as sustainability (Pretty 2008), some argue focusing on clearly defined, measurable outputs or outcomes (Musumba et al. 2017). However, the lessons from human health indicate that frameworks do not only aid policy development and tactical decision-making but also jointly guide application of practices and support communication with the larger audience (CCSDH 2015).

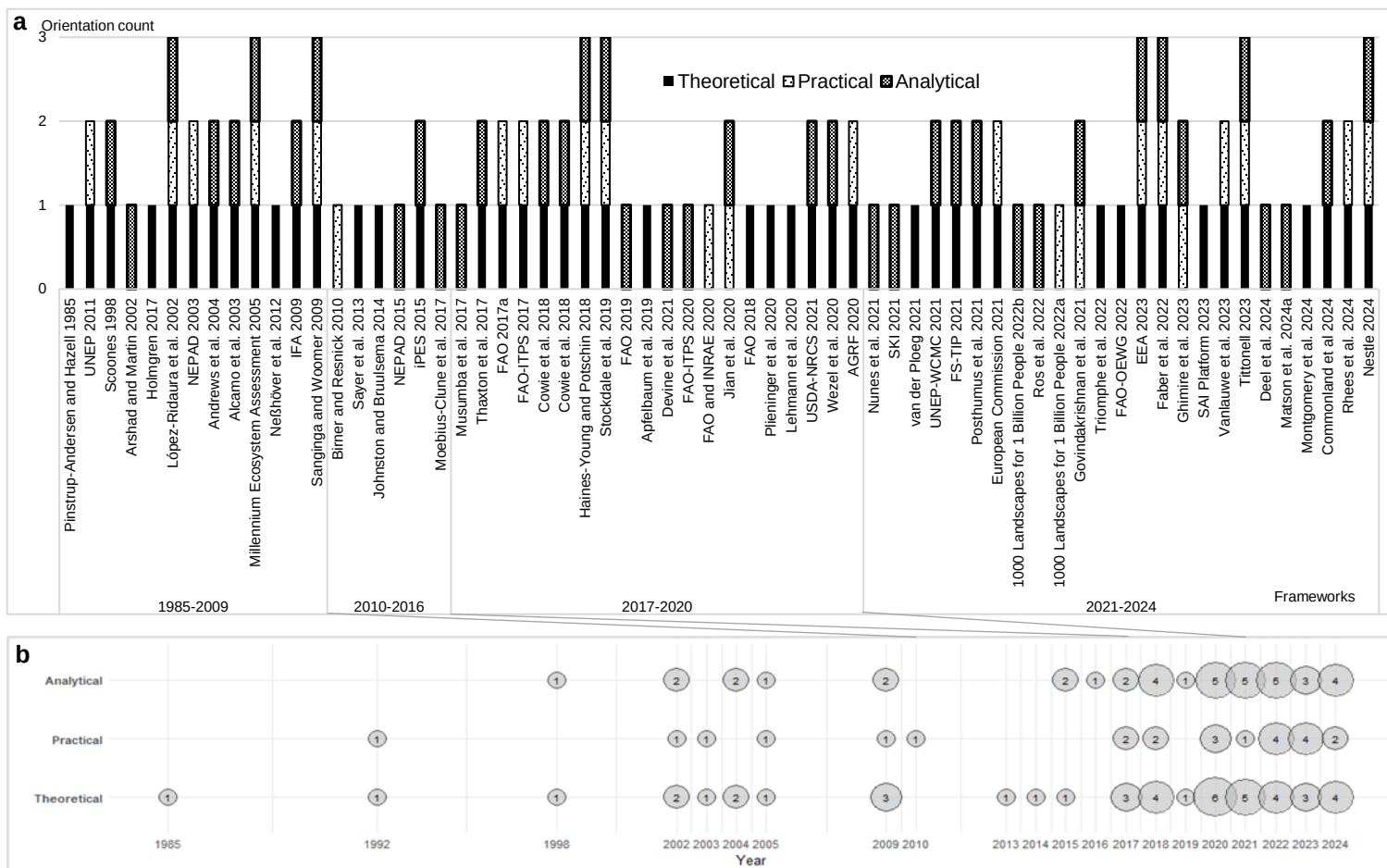


Figure 2 Use-orientation of reviewed frameworks over time.

(a) Frameworks ($n = 64$) ordered chronologically by publication year and classified into three use-orientation categories, (b) Distribution of use-orientation categories by year; circle size reflects the number of frameworks published in that year within each category.

Notably, some organizations develop and/or deploy a set of frameworks depending on use, shifting emphasis over time. For instance, the African Union's Comprehensive Africa Agriculture Development Programme (CAADP) developed in 2003 (NEPAD 2003) focused on theory and practice but shifted towards measuring impact with the 2015 CAADP results framework (NEPAD 2015). This temporal pattern reveals four waves in the evolution of framework use-orientation (Figure 2.b).

The first wave (1985–2009) established early conceptual foundations around soil-quality and efficiency (Arshad and Martin 2002; Andrews et al. 2004) along with ecosystem service framing (Alcamo et al. 2003; Millennium Ecosystem Assessment 2005) and soil fertility management guidance (Sanginga and Woomer 2009). The second wave (2010–2016) marked a landscape/food-systems transition, shifting from fields to territories with landscape and political economy (Birner and Resnick 2010; Sayer et al. 2013; van der Ploeg 2021). The third wave (2017–2020) reflects growing demand for comparable indicators by formalizing agroecology and landscape guidance (FAO 2017a, 2018, 2019), redefining restoration targets using land-

degradation neutrality (Cowie et al. 2018), reframing and synthesis of soil health concepts (Wezel et al. 2020; Lehmann et al. 2020). The fourth wave (2021-2024) emphasized outcome monitoring and corporate alignment, with policy roadmaps and results framework (European Commission 2021; FAO-OEWG 2022), supporting national/continental monitoring guidance (EEA 2023), private-sector buy-in outcome-based tools (Nestle 2024), and expansion of scoring/data infrastructures (Jian et al. 2020; Deel et al. 2024). Peaks in these years align with climate- and biodiversity-goal setting (e.g., SDGs era), pandemic-era resilience debates, and input-price shocks - further reinforcing attention to monitoring, due diligence, and diversified, landscape-scale solutions. The review traces how framework purposes and indicator emphases shift over time, clarifies why no single framework can serve all decision needs and why selection should be aligned with use cases, scales, and transition stages.

3.2 Agroecological principle-orientation

Hierarchical clustering (Figure 3) groups the agroecological principles and frameworks, the dynamic clustering utilizes presence and frequency to determine optimal branching and convergence. Although indicators are not mutually exclusive and vary in relevance as well as clarity and reliability. The column-level clustering further reveals internal branching/convergence structure among agroecological principles within the biophysical and socio-economic clusters.

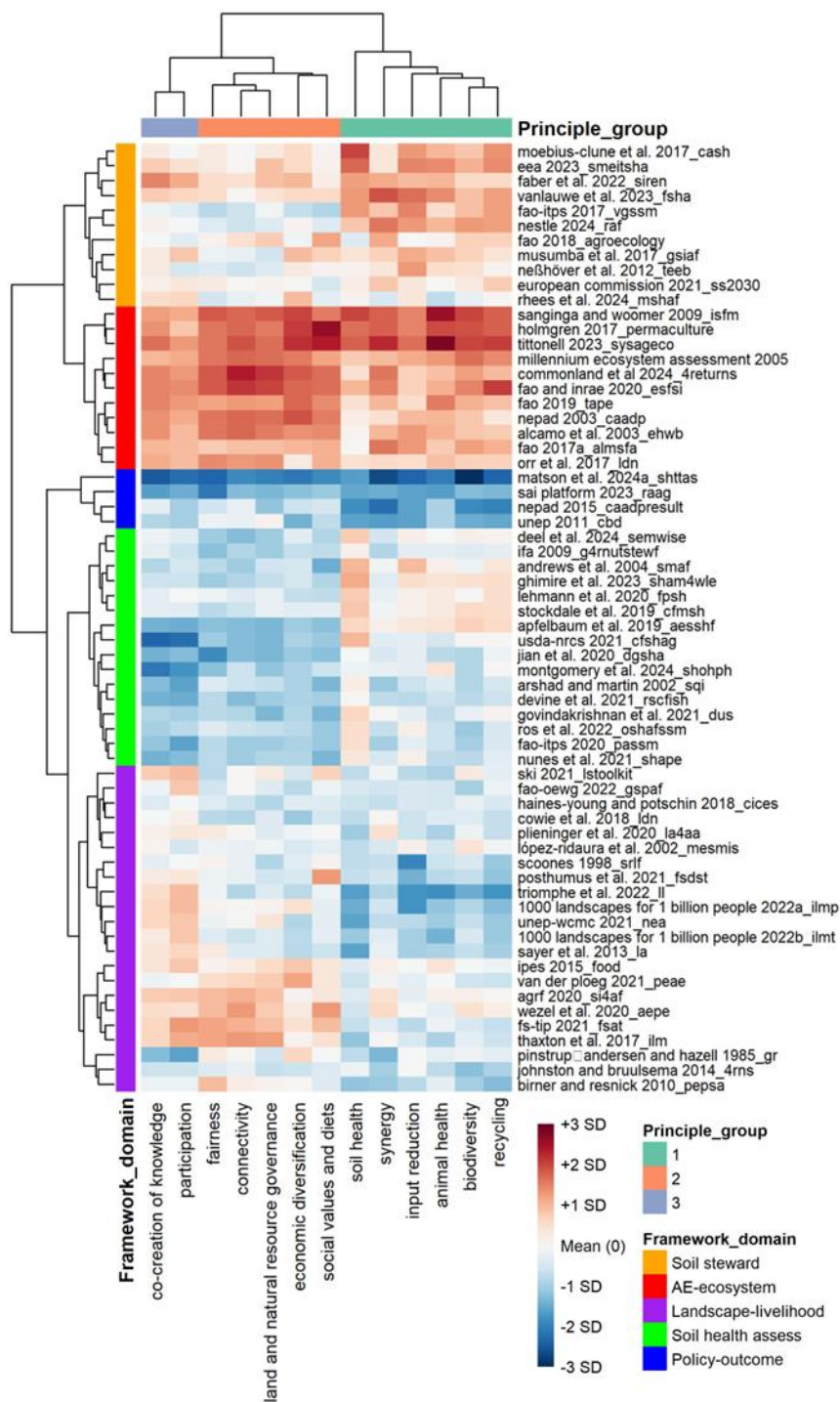


Figure 3 Z-score heatmap of agroecological principle coverage across frameworks ($n = 64$). Rows represent frameworks and columns represent agroecological principles. The color gradient shows z-scores of indicators counts at the principle level, where 0 indicates the mean, positive values indicate above-average coverage up to 3 standard deviations (SD), and negative values indicate below-average coverage. Framework clusters and principle groups are delineated from the dendrogram using Dynamic Tree Cut method. Abbreviated suffixes appended after the author–year citation are used solely for ease of reference and do not represent formal acronyms.

On the biophysical cluster, the hierarchical clustering yields five sub-branches (i) recycling and biodiversity being closely related, then (ii) animal health, (iii) input reduction, (iv) synergy and (v) soil health. This branching contrasts with Wezel et al., (2020), who aggregated the biophysical principles into two pathways (resource efficiency vs resilience). The dendrogram indicates finer, non-binary structure in which several principles form stable micro-constellations rather than collapsing neatly into the two macro-pathways: recycling and biodiversity share lower branching consistent with resilience pathway, while animal health, input reduction and synergy branch at incrementally higher node heights – constructed more like cross-cutting enablers standalone principles.

On the socio-economic cluster: first order constructs aggregate into three pathways: (i) co-creation of knowledge and participation, (ii) fairness, connectivity, and land and natural resource governance, and (iii) economic diversification, and social values and diets. Pathways (ii) and (iii) merging at higher nodes may indicate a common governance-market orientation. The study posits that biophysical convergence reflects the historical evolution of evaluation paradigms. Following a prolonged period of productivism, yield-centric evaluation, “second wave” frameworks standardized field-level soil and agronomic metrics - soil organic matter, erosion control, nutrient use - because they were relatively tractable to measure and had clear plot-scale causal links (Andrews et al. 2004). Socio-economic dimensions were integrated later as attention shifted from plots to farms, landscapes, and food systems, where operationalization required deliberation, institutional arrangements, and context-specific value choices (HLPE 2019; Wezel et al. 2020). Consequently, most frameworks share a common biophysical framing but diverge along a social–economic one.

This asymmetry has direct implications for design, action and monitoring. Since biophysical “indicators” are broadly similar across the assessed frameworks, strategic distinctiveness - and much of the impact variance - arises from socio-economic differences. Effective designs pair the shared biophysical principles with socio-technical contexts: emphasize co-creation where legitimacy and uptake are binding; strengthen fairness, connectivity, and governance rules where coordination and inclusion are weak; and promote diversification, value addition, and diet-quality incentives where demand and profitability are low (Dumont et al. 2021). Understanding the asymmetry, the stakeholders can be guided to: (i) measure what they design and practice: that is, bundling indicators so each practice set is paired with enabling-environment metrics for scaling; (ii) keep plot-scale agronomic indicators to capture short-run change, and (iii) add governance at farm, landscape, and food-system scales to track sustainability, diffusion, and equity.

4 Design domains

The row clustering of Figure 3 delineates five framework design domains – (i) soil management and input stewardship, (ii) soil health assessment, (iii) agroecological and ecosystem-based approaches, (iv) integrated landscape and livelihoods, and (v) policy and outcome-oriented. These five design domains translate complex clustering results into an interpretable typology of how frameworks operationalize soil

health within agri-food systems' transitions, supporting clearer selection, combination, and sequencing for planning and evaluation.

4.1 Soil management and input stewardship

A cluster of frameworks focuses on farm-level resource efficiency, providing operational guidance and measurement routines - soil health practice guidelines (SOM maintenance, pH or salinity correction, compaction avoidance, infiltration, cover and erosion control) are linked to threshold scoring approaches that make outcomes measurable (FAO-ITPS 2017; Moebius-Clune et al. 2017). Public guidance and standards reinforce monitoring for sustainable soil management (Faber et al. 2022; EEA 2023), with broader strategies framing soil protection to meet regional policy targets (European Commission 2021). Within farms, nutrient stewardship emphasizes right source, right rate, right time and right place (4R) decisions and integrated organic inputs to raise uptake while curbing losses and protecting water quality (Musumba et al. 2017; Vanlauwe et al. 2023), while private-sector frameworks align incentives toward restoring SOM, improving nutrient cycling, and lowering synthetic inputs, with co-benefits for climate resilience and farmer livelihoods (Nestle 2024).

Although soil management and input stewardship frameworks supply indicators to verify efficiency gains, they place less emphasis on the enabling conditions: biodiversity and recycling typically enter as due-diligence items or co-benefits of practice (rotations, cover, and organic amendments) rather than primary design features (Neßhöver et al. 2012). Fairness, connectivity, and animal health appear where standards or reporting instruments are included (FAO-ITPS 2017; Moebius-Clune et al. 2017), but co-creation, participation, and diet or consumption elements are comparatively muted across this stewardship-oriented set. Emphasis remains at micro level; landscape hydrology and cross-farm coordination are rarely integral, limiting their capacity to sustain soil health outcomes beyond plot or farm scale, and require complementing tools to address adoption constraints.

4.2 Soil-Health Assessment

The soil-health assessment frameworks are mixed and bimodal. One pole comprises agronomic toolkits that center on soil health, input efficiency, and management synergies. These include indicator suites, scoring systems, and field protocols that translate the soil triad into operational diagnostics (IFA 2009; Stockdale et al. 2019; Lehmann et al. 2020) - biology (active and total carbon, enzymes, microbial biomass), chemistry (pH, macro- or micro-nutrients), and physics (bulk density, penetration resistance, aggregate stability, available water capacity) - with scoring curves, critical limits, or minimum data sets to flag constraints and guide remediation (Arshad and Martin 2002; Andrews et al. 2004; FAO-ITPS 2020; USDA-NRCS 2021; Govindakrishnan et al. 2021; Deel et al. 2024).

The other pole takes social-ecological and health framing, focusing on fairness or participation, connectivity, and diversification while treating biophysical metrics as contextual. Regional or context-specific frameworks

and open databases enable adaptation monitoring and evaluation (Jian et al. 2020; Devine et al. 2021; Ros et al. 2022), with variants focused on water-limited environments and soil organic carbon (Nunes et al. 2021; Ghimire et al. 2023). Recent studies position soil health within one-health or planetary-health agendas, broadening the purpose of diagnostics beyond agronomy (Lehmann et al. 2020; Montgomery et al. 2024). Much as assessment frameworks offer a diagnostic spine (protocols, thresholds, scoring, datasets) that can anchor monitoring and adaptive management.

4.3 Agroecological and Ecosystem-Based

Agroecological and ecosystem-based frameworks pivot from plot-level practices to landscape and system governance, emphasizing co-creation or participation, connectivity and tenure or governance, and multifunctional outcomes (FAO 2018). It couples agroecological practice with territorial coordination, ecosystem-service accounting, and participatory performance tools, moving the locus of change from fields to food systems. One subset focuses on landscape articulation of spatial coordination through stakeholder platforms (FAO 2017b, a), restoration and long-term value creation (Commonland 2024), and system-level feedback (Tittonell 2023). The food-systems sub-block locates farm practices within enabling policy and market settings (FAO and INRAE 2020) and within regional development agendas (NEPAD 2003). Another subset consolidates environmental targets to anchor soil and land stewardship in accounting and planning (Cowie et al. 2018), while ecosystem-service framings provide the benefits ledger and trade-off criterion for design and evaluation (Alcamo et al. 2003; Millennium Ecosystem Assessment 2005). Participatory performance tools operationalize co-creation and learning - from plot to biomes - through context-specific diagnostics and indicators (FAO 2019).

Relative to management and stewardship frameworks, soil functions are secured indirectly - through diversity, ground cover, water regulation, and circular nutrient flows. Agroecological and ecosystem-service approaches promote permanent cover, intercropping, N-fixing species, integrated crop–livestock systems, and reuse of organic resources (Alcamo et al. 2003; Holmgren 2017; FAO 2019). Landscape instruments strengthen erosion control and hydrological regulation via mosaics of land uses, riparian buffers, terraces, and restoration zones, and they institutionalize cross-farm coordination (NEPAD 2003; FAO 2017a; Cowie et al. 2018; Commonland 2024). Within this domain, farm-level nutrient guidance is retained but is explicitly embedded in system design - as one lever among diversification, agroecological redesign, and social organization (Sanginga and Woome 2009; Tittonell 2023). Biodiversity, recycling or circularity, and land and natural-resource governance dominate, while farm-scale input-reduction and within-farm “synergy” register weakly.

Fairness, connectivity, tenure, and coordination are expressed through collective rules and planning (NEPAD 2003; Cowie et al. 2018; FAO and INRAE 2020), co-creation or participation appears where participatory performance tools structure co-learning across scales (FAO 2019), while diet or consumption

elements are secondary and often contingent on food system policies rather than field protocols (FAO and INRAE 2020).

Agroecological and ecosystem-service frameworks embed soil health within territorial governance, coordination, and multifunctionality, shifting the locus of change from field practices to systems organization. Their strength is institutionalizing collective action and landscape integrity, while detailed agronomic routines are typically delegated to complementary toolkits.

4.4 Integrated Landscape and Livelihood

Emerging, this domain groups broad, integrative frameworks that address multiple dimensions (environmental, social, economic) and multiple scales (from farm to landscape to national) in a systems perspective. These frameworks fuse food-system indicator sets with landscape or governance instruments. Agriculture and natural resource management are considered as part of a complex socio-ecological system – factoring in livelihoods, governance, equity, and long-term system change. The unifying characteristic is a holistic view that bridges traditional sectoral silos, aiming for simultaneous progress in productivity, poverty reduction, and environmental sustainability.

Despite the scaling potential, it is worth noting that the geographic coverage and intended applications may vary widely. Frameworks originating in the Global North can encode institutional assumptions distinct from those co-designed in smallholder contexts. Co-creation, participation, fairness, and connectivity are built in via livelihoods, food-systems and political economy lenses (Scoones 1998; López-Ridaura et al. 2002; iPES 2015; Food System Transformative Integrated Policy (FS-TIP) initiative 2021; Posthumus et al. 2021; van der Ploeg 2021; Triomphe et al. 2022). Land- and natural-resource governance is formalized through integrated landscape management and restoration planning (Sayer et al. 2013; Cowie et al. 2018; SKI 2021; 1000 Landscapes for 1 Billion People 2022a, b). Biodiversity and ecosystem-service accounting anchor benefit accounting and monitoring benchmarks (Haines-Young and Potschin-Young 2018; UNEP-WCMC 2021). Diets and economic diversification appear through food-systems analysis and agroecology reviews rather than design features (iPES 2015; Wezel et al. 2020; Food System Transformative Integrated Policy (FS-TIP) initiative 2021). Animal health is sparsely covered (Johnston and Bruulsema 2014; AGRF 2020; FAO-OEWG 2022; Rhees et al. 2024).

Soil outcomes are also secured indirectly - via diversified land use (e.g., agroforestry), ground cover at scale, reduced soil erosion, and improved infiltration - while technical specifics (pH or salinity correction, traffic management, nutrient timing) are typically delegated to complementary agronomic toolkits (Sayer et al. 2013; Plieninger et al. 2020). Integrated frameworks bridge scales and dimensions to support transformation agendas, but their applicability depends on contextual fit and often requires coupling with soil-specific diagnostic and management tools to translate broad objectives into actionable soil stewardship.

4.5 Policy and outcome-orientation

The high-level policy frameworks and outcome-based standards set targets, indicators, or broad principles to guide sustainable agriculture and resource management. They are often driven by governments, international bodies, or industry coalitions and serve as “reference frameworks” for aligning strategies and monitoring progress toward sustainability goals. The common thread is establishing measurable goals or commitments – whether for soil health, biodiversity, or agricultural performance – and creating accountability through indicators and reporting. Policy and convention instruments provide the anchor for tracking global obligations, continental agendas and national commitments while sectoral standards translate these into soil health target or threshold and regenerative outcome metrics (NEPAD 2011; UNEP 2011; SAI Platform 2023; Matson et al. 2024).

Soil health is framed via high-level targets, thresholds, or minimum safeguards (e.g., indicator “floors” for organic carbon, erosion risk, or basic biodiversity). The emphasis is on measurable goals, indicators, and reporting - mainstreaming biodiversity and soil health through key performance indicators - rather than on detailed, farm-level practice and measurement guidelines (Pinstrup-Andersen and Hazell 1985). In this configuration, soil health, input reduction, and biodiversity function mainly as baseline safeguards (e.g., indicator floors); recycling, synergy, co-creation or participation, and consumption-side elements are largely absent. The domain strengthens accountability through targets and comparability yet risks superficial compliance if not linked to diagnostics and practice bundles. The greatest value is the enabling monitoring–reporting–verification through which soil stewardship can be incentivized and aligned across programs.

5 Integration and programming

Content analysis and clustering indicate that frameworks tend to be designed following complementarity, not substitution. Metrics increasingly converge around shared biophysical and agroecological principles, which creates a stronger basis for more coherent and integrated assessment systems (Dumont et al. 2016). Historically, framework development appears to have evolved in response to emerging needs: from an initial emphasis on soil health measurement and management, to landscape-based scaling, and more recently to outcome-, governance- and stewardship-oriented approaches. Rather than representing competing paradigms, the framework clusters can be understood as complementary design domains, each addressing distinct constraints encountered during farming-system transition. Building on these findings, we propose a target-based programming guide that operationalizes complementarity across clusters and domains, positioning soil health as central organizing principle within adaptive, multi-scale transition design (Figure 4).

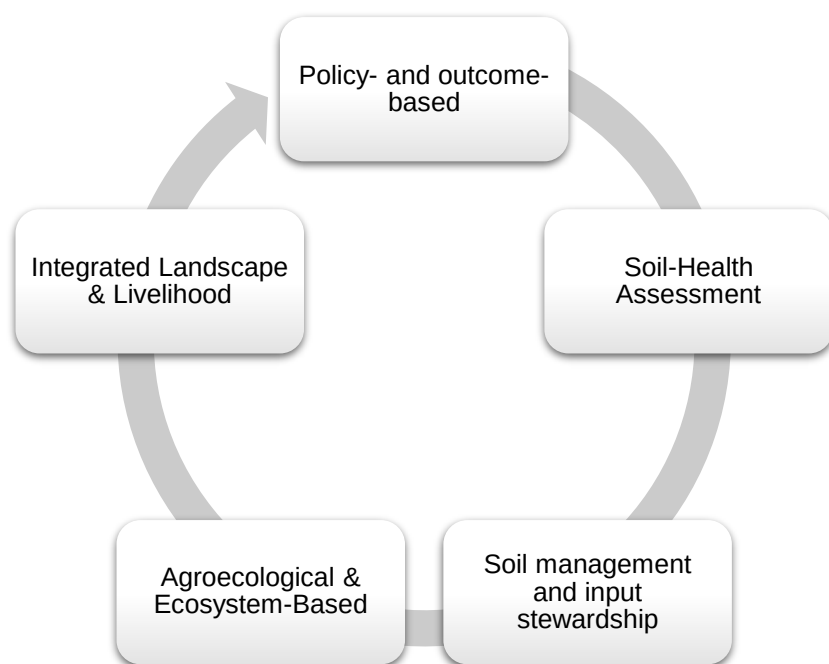


Figure 4 Sequenced integration of the five framework design domains within an adaptive programming cycle to support iterative learning.

The guide is intended to support stepwise programming rather than a one-framework-per-domain selection exercise. For a given intervention, the process begins by defining the purpose of the assessment, the decision context, and the spatial scale. Policy- and outcome-oriented frameworks can be used to set measurable goals, performance criteria, and accountability mechanisms. At this stage, users should identify a limited number of critical points that either weaken or strengthen soil health, drawing on literature review, expert consultation, and participatory diagnosis. This first filtering step is important to avoid an overabundance of metrics and to focus only on the most decision-relevant domains and indicators.

Once priorities are identified, stakeholders can select one or several frameworks within each relevant domain and extract candidate indicators from them. The aim is not to adopt entire frameworks wholesale, but to use them as structured sources of indicators, concepts, and measurement logic. Indicators can then be screened for relevance, feasibility, interpretability, and redundancy, producing a parsimonious set adapted to the intervention context. In this way, the process moves across domains while retaining only those elements that contribute directly to the assessment objective.

Soil health assessment frameworks are particularly useful at this stage for establishing baseline diagnostics and standardized measurement protocols. The methods and tools used to assess soil health and related agroecological indicators should be adapted to local conditions and translated into forms that are meaningful to a wider audience, including farmers, extension agents, and local decision-makers. Soil stewardship and management frameworks can then be used to identify and test practice options and operational routines. Rather than prescribing a fixed intervention package, the compendium and decision

matrix should be used as a basket of options from which context-appropriate combinations can be selected. Field and farm segmentation is important here, because different practices may be suitable for different farming systems, resource endowments, and landscape units.

For example, in a maize-based smallholder system, programming might combine: (i) condition indicators such as soil organic matter, infiltration, and nutrient balance from a Soil health assess framework; (ii) management indicators such as residue retention, crop rotation, or organic amendments from a Soil steward framework; (iii) biodiversity and resilience indicators from an AE-ecosystem framework; and (iv) equity, market access, or governance indicators from Landscape-livelihood and Policy-outcome frameworks. These indicators would then be consolidated into one context-specific monitoring set aligned with the intervention's goals and practical data constraints.

Scaling should then be organized within agroecological and ecosystem boundaries so that adoption does not undermine biophysical integrity or intensify trade-offs as interventions expand. Local scaling partners play a central role in co-designing this strategy. Because landscape boundaries are nested, overlapping, and socially differentiated, integrated programming requires frameworks that address both ecological and social dimensions. The compendium therefore supports not only biophysical targeting, but also the integration of governance, market linkages, and livelihood objectives.

Although Figure 4 presents a sequenced learning cycle, the process is not strictly linear. Spatial scales are nested or overlapping, and feedbacks occur through shorter iterative loops of learning, course correction, and adaptation. Assessing the relevance, feasibility, and accuracy of selected metrics and indicators should therefore be part of the adaptive learning process itself. By reframing frameworks as complementary design domains linked through an adaptive programming cycle, research-for-development can operationalize soil health as a systems-level organizing principle rather than as a stand-alone assessment category.

6 Conclusion

Soil health is widely recognized as an important biophysical and socio-economic dimension of agroecological and agri-food system transitions, yet the proliferation of frameworks has contributed to fragmented guidance and uneven integration across assessment and design efforts. This review offers a comparative structure for examining heterogeneous framework families and for considering how institutions position soil-related indicators within broader system priorities. Across the compendium, variation among frameworks appears to reflect not only methodological differences, but also institutional strategies shaped by utility considerations and normative orientations that influence problem framing, indicator selection, and the balance between technical performance and social legitimacy.

The analysis suggests that many frameworks converge around a set of shared biophysical principles, while showing greater variation in socio-economic dimensions. This pattern may indicate that differences in governance, inclusion, learning processes, and market conditions are particularly important in shaping how

soil health is interpreted and operationalized in practice. The five design domains proposed here translate these patterns into a working typology. Stewardship and assessment domains tend to support practice guidance and diagnostic routines; agroecological or ecosystem, and integrated landscape or livelihood, domains help situate scaling within territorial and governance contexts; and policy or outcome domains can help frame targets, accountability, and comparability across monitoring and reporting systems.

Because these domains appear to address different but related constraints, soil stewardship may be better supported through sequenced combinations of frameworks than through reliance on a single framework alone. On this basis, instead of proposing another standalone framework, the review outlines a programming pathway that begins with policy or outcome targets, links them to soil health diagnostics, uses stewardship-oriented frameworks to inform practice options, considers scaling within agroecological or ecosystem boundaries, extends analysis to landscape and livelihood contexts, and returns to diagnostics through iterative learning. This sequencing is intended as a practical way of using complementarity across frameworks and may help position soil health more centrally within efforts to support resilient, equitable, and sustainable agri-food system transitions.

Acknowledgements

We would like to thank Kate Wellard of McKnight Foundation and Laurie Drinkwater of Cornell University for initiating the discussion on integrated soil health framework, bringing together their expertise and teams on biophysical and political economy to Lilongwe Soil Health Workshop in October of 2024 co-funded by McKnight Foundation and CIMMYT, where this work was structured. We also extend our thanks to all workshop participants from MwAPATA Institute, FIPS Africa, CIMMYT, and LUANAR.

Declarations

Funding

This research was funded by the CGIAR System Council through the Sustainable Farming and the Multi-Functional Landscapes Programs. Author JGM receives salary from MwAPATA Institute, where she is Research Analyst. Partial financial support was received from The McKnight Foundation, which co-financed the Lilongwe Soil Health Workshop.

Conflict of interest

The authors declare no competing interests

Ethics and consent

Not applicable

Availability of data and material

The indicator terms, datasets extracted and/or analyzed during the study are available in the Zenodo repository, <https://zenodo.org/records/17453617>.

Code availability

The R code used for PDF parsing, term proximity extraction, framework-by-principle evidence matrix construction, and comparative framework analysis is available on corresponding author's GitHub repository, https://github.com/powellmponela/soilhealth_ae_framework_review

Authors contributions

Conceptualization, P.M., V.G.P.C., C.K., J.M., R.K. and S.S.; Methodology P.M., V.G.P.C., M.C., and R.K.; Writing Original Draft, P.M. and V.G.P.C.; Writing -Review & Editing, S.L-R., S.S., P.M, V.G.P.C., C.K., J.M., M.C., and R.K.; Funding Acquisition, S.S., S.L-R. and V.G.P.K.; Supervision, S.S.

References

- 1000 Landscapes for 1 Billion People (2022a) Integrated Landscape Management Tool Guide. Companion to “A Practical Guide to Integrated Landscape Management.” https://landscapes.global/wp-content/uploads/2022/10/ILM_Tool_Guide.pdf. Accessed 29 Jul 2025
- 1000 Landscapes for 1 Billion People (2022b) A Practical Guide to Integrated Landscape Management. https://landscapes.global/wp-content/uploads/2022/12/ILM_Practical_Guide_DEC22.pdf. Accessed 29 Jul 2025
- AGRF (2020) Soil Initiative for Africa: Framework Document. https://au.int/sites/default/files/newsevents/workingdocuments/43470-wd-3_EN_Soil_Initiative_for_Africa_Framework_VI_170523.pdf. Accessed 29 Jul 2025
- Alcamo J, Bennett EM, (Program) MEA (2003) Ecosystems and Human Well-being: A Framework for Assessment. Island Press, Washington DC
- Aleskerov F, Bouyssou D, Monjardet B (2007) Utility Maximization, Choice and Preference. Springer Berlin Heidelberg, Berlin: Germany
- Altieri MA, Nicholls CI, Henao A, Lana MA (2015) Agroecology and the design of climate change-resilient farming systems. *Agron Sustain Dev* 35:869–890. <https://doi.org/10.1007/s13593-015-0285-2>
- Amundson R, Berhe AA, Hopmans JW, et al (2015) Soil and human security in the 21st century. *Science* (1979) 348:. <https://doi.org/10.1126/science.1261071>

563 Andrews SS, Karlen DL, Cambardella CA (2004) The Soil Management Assessment Framework. *Soil Sci Soc Am J*
 564 68:1945–1962. [https://doi.org/https://doi.org/10.2136/sssaj2004.1945](https://doi.org/10.2136/sssaj2004.1945)

565 Andrieu N, Dedieu B, Girard P, et al (2025) Methodological challenges in assessing the viability of agroecological
 566 practices: lessons from a multi-case study in Africa. *Agron Sustain Dev* 45:15. [https://doi.org/10.1007/s13593-](https://doi.org/10.1007/s13593-025-01010-9)
 567 025-01010-9

568 Apfelbaum SI, Mcelligott K, Thompson R, Tiller E (2019) AES White Paper: Defining Soil Health Within the Context of
 569 Ecosystem Health-A Framework. [https://uhcc.uh.edu/outreach/practitioners-forum/2022/Steve-Apfelbaum--](https://uhcc.uh.edu/outreach/practitioners-forum/2022/Steve-Apfelbaum--AES-Soil-Health-Framework.pdf)
 570 AES-Soil-Health-Framework.pdf. Accessed 29 Jul 2025

571 Arshad MA, Martin S (2002) Identifying critical limits for soil quality indicators in agro-ecosystems. *Agric Ecosyst*
 572 *Environ* 88:153–160. [https://doi.org/10.1016/S0167-8809\(01\)00252-3](https://doi.org/10.1016/S0167-8809(01)00252-3)

573 Bezner Kerr R, Liebert J, Kansanga M, Kpienbaareh D (2022) Human and social values in agroecology. *Elem Sci Anthr*
 574 10:00090. <https://doi.org/10.1525/elementa.2021.00090>

575 Bicksler AJ, Mottet A, Lucantoni D, et al (2023) The 10 Elements of Agroecology interconnected: Making them
 576 operational in FAO’s work on agroecology. *Elem Sci Anthr* 11:. <https://doi.org/10.1525/elementa.2022.00041>

577 Birner R, Resnick D (2010) The Political Economy of Policies for Smallholder Agriculture. *World Dev* 38:1442–1452.
 578 <https://doi.org/10.1016/j.worlddev.2010.06.001>

579 Cambridge University Press (2025) FRAMEWORK | English meaning - Cambridge Dictionary. In: Cambridge University
 580 Press & Assessment. <https://dictionary.cambridge.org/dictionary/english/framework>. Accessed 27 Oct 2025

581 CCSDH (2015) A Review of Frameworks on the Determinants of Health.
 582 http://ccsdh.ca/images/uploads/Frameworks_Report_English.pdf. Accessed 3 Nov 2024

583 Ceppatelli A, Cremilleux M, Michaud A, Sturaro E (2025) Indicators for animal health on agroecological dairy farms
 584 in mountain areas. *Ital J Anim Sci* 24:2013–2028. <https://doi.org/10.1080/1828051X.2025.2554933>

585 Commonland (2024) The 4 returns framework in practice: a guidebook for holistic landscape restoration.
 586 <https://www.wetlands.org/publication/the-4-returns-framework-in-practice-guidebook/#>. Accessed 29 Jul
 587 2025

588 Cowie AL, Orr BJ, Castillo Sanchez VM, et al (2018) Land in balance: The scientific conceptual framework for Land
 589 Degradation Neutrality. *Environ Sci Policy* 79:25–35. <https://doi.org/10.1016/j.envsci.2017.10.011>

590 Deel HL, Moore JM, Manter DK (2024) SEMWISE: A national soil health scoring framework for agricultural systems.
 591 *Appl Soil Ecol* 195:. <https://doi.org/10.1016/j.apsoil.2024.105273>

Derpsch R (2003) Conservation Tillage, No-Tillage and Related Technologies. In: García-Torres L, Benites J, Martínez-Vilela A, Holgado-Cabrera A (eds) Conservation Agriculture. Springer Netherlands, Dordrecht, pp 181–190

Devine SM, Steenwerth KL, O’Geen AT (2021) A regional soil classification framework to improve soil health diagnosis and management. *Soil Sci Soc Am J* 85:361–378. <https://doi.org/10.1002/saj2.20200>

Dumont AM, Vanloqueren G, Stassart PM, Baret P V. (2016) Clarifying the socioeconomic dimensions of agroecology: between principles and practices. *Agroecol Sustain Food Syst* 40:24–47. <https://doi.org/10.1080/21683565.2015.1089967>

Dumont AM, Wartenberg AC, Baret P V. (2021) Bridging the gap between the agroecological ideal and its implementation into practice. A review. *Agron Sustain Dev* 41:32. <https://doi.org/10.1007/s13593-021-00666-3>

Dumont B, Barlagne C, Cassart P, et al (2024) Principles, barriers and enablers to agroecological animal production systems: a qualitative approach based on five case studies. *Animal*. <https://doi.org/10.1016/j.animal.2024.101367>

EEA (2023) Soil monitoring in Europe: indicators and thresholds for soil quality assessments. <https://www.eea.europa.eu/en/analysis/publications/soil-monitoring-in-europe>. Accessed 29 Jul 2025

European Commission (2021) EU Soil Strategy for 2030: Reaping the benefits of healthy soils for people, food, nature and climate. <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52021DC0699>. Accessed 29 Jul 2025

Faber JH, Hanegraaf M, Gillikin A, et al (2022) Stocktaking for Agricultural Soil Quality and Ecosystem Services Indicators and their Reference Values (SIREN). www.ejpsol.eu. Accessed 3 Oct 2024

FAO (2019) TAPE Tool for Agroecology Performance Evaluation 2019- Process of development and guidelines for application. Test version. <https://openknowledge.fao.org/items/8511c796-c7d1-4a04-895d-a28115731ce0>. Accessed 19 Sep 2024

FAO (2017a) Landscapes for Life: approaches to landscape management for sustainable food and agriculture. <https://openknowledge.fao.org/server/api/core/bitstreams/ff1e6259-a55d-4a79-b239-81d18e243538/content>. Accessed 29 Jul 2025

FAO (2018) The 10 elements of Agroecology: Guiding the transition to sustainable food and agricultural systems. <http://www.fao.org/3/I9037EN/I9037en.pdf>. Accessed 20 Sep 2024

FAO (2017b) Climate-Smart Agriculture Sourcebook. <https://openknowledge.fao.org/handle/20.500.14283/i7994en>. Accessed 12 Nov 2024

622 FAO and INRAE (2020) Enabling sustainable food systems: Innovators' handbook. In: Enabling sustainable food
623 systems. <https://doi.org/10.4060/ca9917en>. Accessed 29 Jul 2025

624 FAO-ITPS (2015) Status of the World's Soil Resources.
625 [https://openknowledge.fao.org/server/api/core/bitstreams/6ec24d75-19bd-4f1f-b1c5-](https://openknowledge.fao.org/server/api/core/bitstreams/6ec24d75-19bd-4f1f-b1c5-5becf50d0871/content)
626 [5becf50d0871/content](https://openknowledge.fao.org/server/api/core/bitstreams/6ec24d75-19bd-4f1f-b1c5-5becf50d0871/content). Accessed 19 Sep 2024

627 FAO-ITPS (2017) Voluntary Guidelines for Sustainable Soil Management.
628 [https://openknowledge.fao.org/server/api/core/bitstreams/9a5b9373-3558-43b3-b732-](https://openknowledge.fao.org/server/api/core/bitstreams/9a5b9373-3558-43b3-b732-f69326a7314d/content)
629 [f69326a7314d/content](https://openknowledge.fao.org/server/api/core/bitstreams/9a5b9373-3558-43b3-b732-f69326a7314d/content). Accessed 29 Jul 2025

630 FAO-ITPS (2020) Protocol for the assessment of Sustainable Soil Management.
631 https://www.fao.org/fileadmin/user_upload//GSP/SSM/SSM_Protocol_EN_006.pdf. Accessed 29 Jul 2025

632 FAO-OEWG (2022) Global Soil Partnership Action Framework 2022-2030 - Healthy soils for a healthy life and
633 environment: from promotion to consolidation of sustainable soil management.
634 [https://openknowledge.fao.org/server/api/core/bitstreams/8428f509-cadf-48c8-845a-](https://openknowledge.fao.org/server/api/core/bitstreams/8428f509-cadf-48c8-845a-58fac410e11e/content)
635 [58fac410e11e/content](https://openknowledge.fao.org/server/api/core/bitstreams/8428f509-cadf-48c8-845a-58fac410e11e/content). Accessed 29 Jul 2025

636 Foley JA, Ramankutty N, Brauman KA, et al (2011) Solutions for a cultivated planet. *Nature* 478:337–342.
637 <https://doi.org/10.1038/nature10452>

638 Food System Transformative Integrated Policy (FS-TIP) initiative (2021) Food Systems Analysis Toolkit A Tool to
639 Accelerate Food Systems Transformation By the Food System Transformative Integrated Policy (FS-TIP)
640 initiative. [https://www.rockefellerfoundation.org/wp-content/uploads/2022/02/Food-Systems-Analysis-](https://www.rockefellerfoundation.org/wp-content/uploads/2022/02/Food-Systems-Analysis-Toolkit.pdf)
641 [Toolkit.pdf](https://www.rockefellerfoundation.org/wp-content/uploads/2022/02/Food-Systems-Analysis-Toolkit.pdf). Accessed 2 Mar 2025

642 Ghimire R, Thapa VR, Acosta-Martinez V, et al (2023) Soil Health Assessment and Management Framework for
643 Water-Limited Environments: Examples from the Great Plains of the USA. *Soil Syst* 7:22.
644 <https://doi.org/10.3390/soilsystems7010022>

645 Giller KenE, Andersson JA, Corbeels M, et al (2015) Beyond conservation agriculture. *Front Plant Sci* 6:.
646 <https://doi.org/10.3389/fpls.2015.00870>

647 Godfray HCJ, Beddington JR, Crute IR, et al (2010) Food Security: The Challenge of Feeding 9 Billion People. *Science*
648 (1979) 327:812–818. <https://doi.org/10.1126/science.1185383>

649 Govindakrishnan PM, Ganeshamurthy AN, Krishna Kumar NK, et al (2021) DUS(Soil)—A Framework for Developing a
650 Minimum Data Set of Soil Health Indicators and Management Guidelines for Farmers. *J Agric Sci* 14:41.
651 <https://doi.org/10.5539/jas.v14n1p41>

652 Greco S, Ishizaka A, Tasiou M, Torrì G (2019) On the Methodological Framework of Composite Indices: A Review of
653 the Issues of Weighting, Aggregation, and Robustness. *Soc Indic Res* 141:61–94.
654 <https://doi.org/10.1007/s11205-017-1832-9>

655 Haines-Young R, Potschin-Young M (2018) Common International Classification of Ecosystem Services (CICES) V5.1-
656 Guidance on the Application of the Revised Structure.
657 https://cices.eu/content/uploads/sites/8/2025/04/CICES-V5.2_Guidance_final_-_29042025.pdf. Accessed 30
658 Sep 2024

659 Hazard L, Crivits M, Ducos A, et al (2026) Pragmatic principles for supporting the agroecological transition in the
660 field: a review. *Agron Sustain Dev* 46:17. <https://doi.org/10.1007/s13593-026-01092-z>

661 HLPE (2019) Agroecological and other innovative approaches for sustainable agriculture and food systems that
662 enhance food security and nutrition. A report by the High Level Panel of Experts on Food Security and Nutrition
663 of the Committee on World Food Security. www.fao.org/cfs/cfs-hlpe. Accessed 24 Sep 2024

664 Holmgren D (2017) Permaculture: Principles & Pathways Beyond Sustainability.
665 <https://au.permacultureprinciples.com/product/principles-and-pathways/>. Accessed 29 Jul 2025

666 IFA (2009) The Global “4R” Nutrient Stewardship Framework: Developing Fertilizer Best Management Practices for
667 Delivering Economic, Social and Environmental Benefits. [https://www.fertilizer.org/wp-](https://www.fertilizer.org/wp-content/uploads/2023/01/2009_ifa_agcom_FBMPs.pdf)
668 [content/uploads/2023/01/2009_ifa_agcom_FBMPs.pdf](https://www.fertilizer.org/wp-content/uploads/2023/01/2009_ifa_agcom_FBMPs.pdf). Accessed 29 Jul 2025

669 IPCC (2019) Climate Change and Land: An IPCC special report. <https://www.ipcc.ch/srccl/>. Accessed 12 Dec 2024

670 IPES (2015) THE NEW SCIENCE OF SUSTAINABLE FOOD SYSTEMS: Overcoming Barriers to Food Systems Reform.
671 https://www.ipes-food.org/_img/upload/files/NewScienceofSusFood.pdf. Accessed 29 Jul 2025

672 Jian J, Du X, Stewart RD (2020) A database for global soil health assessment. *Sci Data* 7:.
673 <https://doi.org/10.1038/s41597-020-0356-3>

674 Johnston AM, Bruulsema TW (2014) 4R Nutrient Stewardship for Improved Nutrient Use Efficiency. *Procedia Eng*
675 83:365–370. <https://doi.org/10.1016/j.proeng.2014.09.029>

676 Kibblewhite MG, Ritz K, Swift MJ (2008) Soil health in agricultural systems. *Philos Trans R Soc B Biol Sci* 363:685–701.
677 <https://doi.org/10.1098/rstb.2007.2178>

678 Lal R (2004) Soil Carbon Sequestration Impacts on Global Climate Change and Food Security. *Science* (1979)
679 304:1623–1627. <https://doi.org/10.1126/science.1097396>

680 Langfelder P, Zhang B, Horvath S (2008) Defining clusters from a hierarchical cluster tree: the Dynamic Tree Cut
681 package for R. *Bioinformatics* 24:719–720. <https://doi.org/10.1093/bioinformatics/btm563>

682 Lehmann J, Bossio DA, Kögel-Knabner I, Rillig MC (2020) The concept and future prospects of soil health. *Nat Rev*
 683 *Earth Environ* 1:544–553. <https://doi.org/10.1038/s43017-020-0080-8>
 684 López-Ridaura S, Masera O, Astier M (2002) Evaluating the sustainability of complex socio-environmental systems.
 685 The MESMIS framework. *Ecol Indic* 2:135–148. [https://doi.org/10.1016/S1470-160X\(02\)00043-2](https://doi.org/10.1016/S1470-160X(02)00043-2)
 686 Lucantoni D, Sy MR, Goïta M, et al (2023) Evidence on the multidimensional performance of agroecology in Mali
 687 using TAPE. *Agric Syst* 204:103499. <https://doi.org/10.1016/j.agsy.2022.103499>
 688 Luft JA, Jeong S, Idsardi R, Gardner G (2022) Literature Reviews, Theoretical Frameworks, and Conceptual
 689 Frameworks: An Introduction for New Biology Education Researchers. *CBE Life Sci Educ* 21:rm33.
 690 <https://doi.org/10.1187/cbe.21-05-0134>
 691 MacKenzie SB, Podsakoff PM, Podsakoff NP (2011) Construct measurement and validation procedures in MIS and
 692 behavioral research: Integrating new and existing techniques. *MIS Q* 35:293–334.
 693 <https://doi.org/10.2307/23044045>
 694 Madsen S, Kerr RB, Kamilia K, et al (2025) Agroecology supports sustainable development in Africa. A review. *Agron*
 695 *Sustain Dev* 45:34. <https://doi.org/10.1007/s13593-024-00976-2>
 696 Matson A, Fantappiè M, Campbell GA, et al (2024) A framework for setting soil health targets and thresholds in
 697 agricultural soils. <https://meetingorganizer.copernicus.org/EGU24/EGU24-20664.html>. Accessed 29 Sep 2024
 698 Mazziotta M, Pareto A (2016) On a Generalized Non-compensatory Composite Index for Measuring Socio-economic
 699 Phenomena. *Soc Indic Res* 127:983–1003. <https://doi.org/10.1007/s11205-015-0998-2>
 700 Millennium Ecosystem Assessment (2005) Ecosystems and human well-being : synthesis : a report of the Millennium
 701 Ecosystem Assessment. <https://www.millenniumassessment.org/documents/document.356.aspx.pdf>.
 702 Accessed 29 Jul 2025
 703 Minasny B, Malone BP, McBratney AB, et al (2017) Soil carbon 4 per mille. *Geoderma* 292:59–86.
 704 <https://doi.org/10.1016/j.geoderma.2017.01.002>
 705 Mockshell J, Omulo G, Hidalgo F, et al (2026) Key narratives around agroecological transitions. A systematic literature
 706 review of the current debate. *Agroecol Sustain Food Syst* 50:424–449.
 707 <https://doi.org/10.1080/21683565.2025.2524730>
 708 Moebius-Clune BN, Moebius-Clune DJ, Gugino BK, Idowu O (2017) Comprehensive Assessment of Soil Health - The
 709 Cornell Framework. <https://www.css.cornell.edu/extension/soil-health/manual.pdf>. Accessed 19 Sep 2024
 710 Montgomery DR, Rabinowitz P, Sipos Y, Wheat EE (2024) Soil health: A common focus for one health and planetary
 711 health interventions. *One Health* 18:100673. <https://doi.org/10.1016/j.onehlt.2023.100673>

- Mottet A, Bicksler A, Lucantoni D, et al (2020) Assessing Transitions to Sustainable Agricultural and Food Systems: A Tool for Agroecology Performance Evaluation (TAPE). *Front Sustain Food Syst* 4:. <https://doi.org/10.3389/fsufs.2020.579154>
- Moullin JC, Dickson KS, Stadnick NA, et al (2020) Ten recommendations for using implementation frameworks in research and practice. *Implement Sci Commun* 1:42. <https://doi.org/10.1186/s43058-020-00023-7>
- Musumba M, Grabowski P, Palm C, Snapp S (2017) Guide for the Sustainable Intensification Assessment Framework Acknowledgments. <https://hdl.handle.net/10568/90523>. Accessed 29 Jul 2025
- Nardo M, Saisana M, Saltelli A, et al (2005) Handbook on Constructing Composite Indicators. https://www.oecd.org/en/publications/handbook-on-constructing-composite-indicators_533411815016.html. Accessed 12 Dec 2025
- NEPAD (2003) Comprehensive Africa Agriculture Development Programme. <https://www.nepad.org/file-download/download/public/15904>. Accessed 19 Sep 2024
- NEPAD (2015) The CAADP Results Framework 2015-2025. <https://www.nepad.org/file-download/download/public/14299>. Accessed 19 Sep 2024
- NEPAD (2011) The Abuja Declaration on Fertilizers for an African Green Revolution - Status of Implementation at Regional and National Levels. African Union
- Neßhöver C, James Aronson, James Blignaut, et al (2012) The Economics of Ecosystems and Biodiversity in National and International Policy Making. Routledge
- Nestle (2024) The Nestle Agriculture Framework. <https://www.nestle.com/sites/default/files/2022-07/nestle-agriculture-framework.pdf>. Accessed 29 Jul 2025
- Nicholls CI, Altieri MA, Kobayashi M, et al (2020) Assessing the agroecological status of a farm: a principle-based assessment tool for farmers. *Agro Sur* 48:29–41. <https://doi.org/10.4206/agrosur.2020.v48n2-04>
- Nunes MR, Veum KS, Parker PA, et al (2021) The soil health assessment protocol and evaluation applied to soil organic carbon. *Soil Sci Soc Am J* 85:1196–1213. <https://doi.org/10.1002/saj2.20244>
- Oreg S, Katz-Gerro T (2006) Predicting proenvironmental behavior cross-nationally: Values, the theory of planned behavior, and value-belief-norm theory. *Environ Behav* 38:462–483. <https://doi.org/10.1177/0013916505286012>
- Ostrom E (2011) Background on the Institutional Analysis and Development Framework. *Policy Stud J* 39:7–27. <https://doi.org/10.1111/j.1541-0072.2010.00394.x>

741 Pinstруп-Andersen P, Hazell PBR (1985) The impact of the green revolution and prospects for the future. *Food Rev*
742 *Int* 1:1–25. <https://doi.org/10.1080/87559128509540765>

743 Plieninger T, Muñoz-Rojas J, Buck LE, Scherr SJ (2020) Agroforestry for sustainable landscape management. *Sustain*
744 *Sci* 15:1255–1266. <https://doi.org/10.1007/s11625-020-00836-4>

745 Posthumus H, Bosselaar J, Brouwer H (2021) The Food Systems Decision-Support Toolbox- a toolbox for food system
746 analysis. <https://edepot.wur.nl/541410>. Accessed 19 Sep 2024

747 Pretty J (2008) Agricultural sustainability: concepts, principles and evidence. *Philos Trans R Soc B Biol Sci* 363:447–
748 465. <https://doi.org/10.1098/rstb.2007.2163>

749 Rhees S, Lewandowski AM, Cates A, et al (2024) Minnesota Soil Health Action Framework.
750 <https://hdl.handle.net/11299/261252>. Accessed 29 Jul 2025

751 Rocco ST, Plakhotnik SM (2009) Literature reviews, conceptual frameworks, and theoretical frameworks: Terms,
752 functions, and distinctions. *Human Resource Development Review* 8:120–130.
753 <https://doi.org/10.1177/1534484309332617>

754 Rockström J, Steffen W, Noone K, et al (2009) A safe operating space for humanity. *Nature* 461:472–475.
755 <https://doi.org/10.1038/461472a>

756 Ros GH, Verweij SE, Janssen SJC, et al (2022) An Open Soil Health Assessment Framework Facilitating Sustainable
757 Soil Management. *Environ Sci Technol* 56:17375–17384. <https://doi.org/10.1021/acs.est.2c04516>

758 SAI Platform (2023) Regenerating Together: A global framework for regenerative agriculture.
759 [https://saipatform.org/wp-content/uploads/2023/09/sai-platform_-regenerating-together_september-](https://saipatform.org/wp-content/uploads/2023/09/sai-platform_-regenerating-together_september-2023.pdf)
760 2023.pdf. Accessed 12 Dec 2024

761 Sanchez PA (2002) Soil Fertility and Hunger in Africa. *Science* (1979) 295:2019–2020.
762 <https://doi.org/10.1126/science.1065256>

763 Sanginga N, Woomer PL (2009) Integrated Soil Fertility Management in Africa: Principles Practices and Development
764 Process. In: *Tropical Soil Biology and Fertility Institute of the International Centre for Tropical Agriculture*.
765 Nairobi. http://ciat-library.ciat.cgiar.org/articulos_ciat/tsbf/pdf/tsbf_isfm_book09_contents.pdf. Accessed 19
766 Sep 2024

767 Sayer J, Sunderland T, Ghazoul J, et al (2013) Ten principles for a landscape approach to reconciling agriculture,
768 conservation, and other competing land uses. *Proc Natl Acad Sci* 110:8349–8356.
769 <https://doi.org/10.1073/pnas.1210595110>

770 Scoones I (1998) Sustainable rural livelihoods: a framework for analysis.
 771 <https://www.ids.ac.uk/publications/sustainable-rural-livelihoods-a-framework-for-analysis/>. Accessed 19 Sep
 772 2024

773 SKI (2021) Agroecology at a landscape level - Toolkit for facilitators. [https://www.seedandknowledge.org/wp-](https://www.seedandknowledge.org/wp-content/uploads/2022/12/SKI-AE-Landscape-Toolkit.pdf)
 774 [content/uploads/2022/12/SKI-AE-Landscape-Toolkit.pdf](https://www.seedandknowledge.org/wp-content/uploads/2022/12/SKI-AE-Landscape-Toolkit.pdf). Accessed 29 Jul 2025

775 Stockdale EA, Griffiths BS, Hargreaves PR, et al (2019) Conceptual framework underpinning management of soil
 776 health—supporting site-specific delivery of sustainable agro-ecosystems. *Food Energy Secur* 8:.
 777 <https://doi.org/10.1002/fes3.158>

778 Thierfelder C, Baudron F, Setimela P, et al (2018) Complementary practices supporting conservation agriculture in
 779 southern Africa. A review. *Agron Sustain Dev* 38:16. <https://doi.org/10.1007/s13593-018-0492-8>

780 Tittonell P (2014) Ecological intensification of agriculture—sustainable by nature. *Curr Opin Environ Sustain* 8:53–
 781 61. <https://doi.org/10.1016/j.cosust.2014.08.006>

782 Tittonell P (2023) *A Systems Approach to Agroecology*. Springer Nature Switzerland, Cham

783 Triomphe B, Bergamini N, Fuchs LE (2022) Engaging with stakeholders for initiating agroecological transition in Living
 784 Landscapes Six guiding principles On behalf of the WP1 team CGIAR Initiative on Agroecology.
 785 <https://hdl.handle.net/10568/127414>. Accessed 29 Jul 2025

786 UNEP (2011) Convention on Biological Diversity -text and annexes. <https://www.cbd.int/doc/legal/cbd-en.pdf>.
 787 Accessed 29 Jul 2025

788 UNEP-WCMC (2021) National ecosystem assessments to support implementation of the convention on biological
 789 diversity. <https://www.ecosystemassessments.net/content/uploads/2021/11/CBD-NEA-2021.pdf>. Accessed
 790 29 Jul 2025

791 UNHCR (2017) Needs Assessment Handbook.
 792 <https://www.unhcr.org/handbooks/assessment/sites/assessment/files/2023-10/UNHCR>. Accessed 29 Oct
 793 2024

794 USDA-NRCS (2021) Soil Health Technical Note No. 450-06 Cropland In-Field Soil Health Assessment Guide.
 795 [https://www.nrcs.usda.gov/sites/default/files/2023-03/In-Field Assessment 2021 Complete ND.pdf](https://www.nrcs.usda.gov/sites/default/files/2023-03/In-Field%20Assessment%202021%20Complete%20ND.pdf). Accessed
 796 19 Sep 2024

797 Utter A, White A, Méndez VE, Morris K (2021) Co-creation of knowledge in agroecology. *Elem Sci Anthr* 9:00026.
 798 <https://doi.org/10.1525/elementa.2021.00026>

799 van der Ploeg JD (2021) The political economy of agroecology. *J Peasant Stud* 48:274–297.
800 <https://doi.org/10.1080/03066150.2020.1725489>

801 Vanlauwe B, Amede T, Bationo A, et al (2023) Fertilizer and Soil Health in Africa: The Role of Fertilizer in Building Soil
802 Health to Sustain Farming and Address Climate Change. [https://ifdc.org/wp-](https://ifdc.org/wp-content/uploads/2023/02/Fertilizer-and-Soil-Health-in-Africa-The-Role-of-Fertilizer-in-Building-Soil-Health-to-Sustain-Farming-and-Address-Climate-Change.pdf)
803 [content/uploads/2023/02/Fertilizer-and-Soil-Health-in-Africa-The-Role-of-Fertilizer-in-Building-Soil-Health-](https://ifdc.org/wp-content/uploads/2023/02/Fertilizer-and-Soil-Health-in-Africa-The-Role-of-Fertilizer-in-Building-Soil-Health-to-Sustain-Farming-and-Address-Climate-Change.pdf)
804 [to-Sustain-Farming-and-Address-Climate-Change.pdf](https://ifdc.org/wp-content/uploads/2023/02/Fertilizer-and-Soil-Health-in-Africa-The-Role-of-Fertilizer-in-Building-Soil-Health-to-Sustain-Farming-and-Address-Climate-Change.pdf). Accessed 29 Jul 2025

805 Vanlauwe B, Wendt J, Giller KE, et al (2014) A fourth principle is required to define Conservation Agriculture in sub-
806 Saharan Africa: The appropriate use of fertilizer to enhance crop productivity. *Field Crops Res* 155:10–13.
807 <https://doi.org/10.1016/j.fcr.2013.10.002>

808 Wezel A, Herren BG, Kerr RB, et al (2020) Agroecological principles and elements and their implications for
809 transitioning to sustainable food systems. A review. *Agron Sustain Dev* 40:. [https://doi.org/10.1007/s13593-](https://doi.org/10.1007/s13593-020-00646-z)
810 [020-00646-z](https://doi.org/10.1007/s13593-020-00646-z)

811

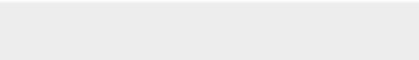
812



Click here to access/download

Manuscript with Track Changes

Out 6. Integrated framework R2 with tracked
changes.docx



Managing Editor:

Please make sure that the revised version of your manuscript adheres to our Instructions for Authors: <https://link.springer.com/journal/13593/submission-guidelines>

For the figures, in particular for Fig. 2a. The title of the Y-axis needs to be written horizontally at the top left corner of the graph, and the legend shall be completed (meaning of the 3 types of bars to be provided, with larger symbols) / reformatted to avoid having words split on two lines. In addition, arial or helvetica (sans serif fonts) shall be used in the whole Fig. 2a. For both Fig. 2a and 2b, the position of (a) and (b) shall be at the top, to the right of the Y-axes / top of the graph. Photo credits must be explicitly mentioned for Fig. 1

Contrary to what is claimed, the list of references is not in line with our guidelines (see instructions for authors). Revise them all carefully.

Actions: We have revised the figures to conform more closely to the journal style. In Figure 2a, the y-axis title has been repositioned and reformatted, the legend has been clarified and enlarged, and the figure has been revised throughout using Arial/Helvetica sans serif fonts. For Figures 2a and 2b, the panel labels have been repositioned as requested. We have also added explicit photo credits to Figure 1. In addition, the full reference list has been carefully checked and revised to align with the journal's Instructions for Authors.

Associate Editor:

A thorough revision was carried out, and the paper is now much clearer and well structured. However, there are still some small corrections and one point regarding the Integration and Programing which merits a bit more work.

Line 174. Delete "B." before Dumont

Action: We changed the reference style to ASD from APA 7 in Mendeley cite.

Figure 2A. We do not see the legend for the analytical dimension

Action: We updated the fill and border of bars, now its distinct and visible.

Figure 3. Van der Ploeg et al. 2021 appears, but has been deleted from the reference list. And you still have Wezel et al. 2020 with TAPE. This is not a TAPE publication. What is fs-tip 2021? Check again the authors of the publication and other terms you use here.

Action: Wezel's publication '*Agroecological principles and elements ...*' has been abbreviated as *aepe*. We have also included a note that the abbreviations should be used for easy reference and do not indicate formal acronyms.

Not clear what are the above mean or below mean numbers. Mean (0) is the mean number of principles covered by all the frameworks? Or is it the mean of the 13 principles, so equivalent to 6.5? And you compare the frameworks to this reference value?

Action: That you for this observation. It would indeed be confusing. We have clarified in the legend that "*The color gradient shows z-scores of indicators counts at the principle level, where 0 indicates the mean, positive values indicate above-average coverage, and negative values indicate below-average coverage*".

Line 309 and 441. Delete A.M. with the Dumont reference

Line 432. Correct reference citing of: (Pinstrup-Andersen Peter R Hazell, 1985)

Figure 4. Delete 1. in the box bottom left. Bottom right start with S and not s

Action: corrected

Section 5 Integration and programing. It gives general guidance how to do this, but it remains the operational question how to do it. Is the recommendation to take from each

domain one framework, or several, and from them select indicators? And then go to the next domain? Not clear at all I would say. For that the paper become more attractive, could you give an example. Eg. Without being here a bit clearer, the reader remains quite frustrated at the end, as he/she does not know how to apply your programming.

Action: Thank you for this important observation. In the revision, we clarify that the approach is not limited to selecting one framework per domain; rather, users may draw on one or several frameworks within each domain, depending on the assessment objective and context, and then extract and harmonize a parsimonious set of indicators before proceeding through the sequence. We have also added a concrete example to illustrate how this programming can be implemented step by step across the five domains.